Applications

Alexander S. Kulikov

Steklov Mathematical Institute at St. Petersburg, Russian Academy of Sciences
and
University of California, San Diego

Outline

Least Common Multiple

Diophantine Equations: Examples

Diophantine Equations: Theorem

Modular Division

Least Common Multiple

Definition

The least common multiple, lcm(a, b), of integers a and b (both different from zero) is the smallest positive integer that is divisible by both a and b

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Examples

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Convention

We assume that a and b are positive

Working with Fractions

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$$\frac{31}{177} + \frac{29}{118} = \frac{31 \cdot 2}{177 \cdot 2} + \frac{29 \cdot 3}{118 \cdot 3}$$

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$$\operatorname{lcm}(177, 118) = 354 = 2 \cdot 177 = 3 \cdot 118$$

$$\frac{31}{177} + \frac{29}{118} = \frac{31 \cdot 2}{177 \cdot 2} + \frac{29 \cdot 3}{118 \cdot 3} = \frac{149}{354}$$

Naive Algorithm

Clearly, $a \cdot b$ is divisible by a and b. To find the *least* common multiple, simply try all numbers up to $a \cdot b$ and select the smallest one

Naive Algorithm: Code

```
def lcm(a, b):
  assert a > 0 and b > 0

  for d in range(1, a * b + 1):
    if d % a == 0 and d % b == 0:
      return d
```

Naive Algorithm: Code

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def lcm(a, b):
  assert a > 0 and b > 0

  for d in range(1, a * b + 1):
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```

```
print(lcm(24, 16))
```

```
48
```

Naive Algorithm: Analysis

- If $\text{lcm}(a, b) = a \cdot b$, the algorithm will perform $a \cdot b$ divisions

Naive Algorithm: Analysis

- If $\text{lcm}(a, b) = a \cdot b$, the algorithm will perform $a \cdot b$ divisions
- Again, very slow: the call

```
print(lcm(531441, 262144))
```

  will take more than one minute

Euclid's algorithm

Can we use efficient Euclid's algorithm to compute the least common multiple, too?



Euclid's algorithm

Can we use efficient Euclid's algorithm to compute the least common multiple, too?

Yes!

$$\operatorname{lcm}(a, b) \cdot \gcd(a, b) = a \cdot b$$



lcm and gcd

Lemma

If $a, b > 0$, then $\mathrm{lcm}(a, b) = ab/\gcd(a, b)$

lcm and gcd

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If $a, b > 0$, then $\text{lcm}(a, b) = ab/\text{gcd}(a, b)$

Proof

- Let $d = \text{gcd}(a, b)$, $a = dp$, $b = dq$

□

lcm and gcd

Lemma

If $a, b > 0$, then $\text{lcm}(a, b) = ab/\gcd(a, b)$

Proof

- Let $d = \gcd(a, b)$, $a = dp$, $b = dq$
- Then $m = ab/d = pb = qa$ is a multiple of a and b

□

lcm and gcd

Lemma

If $a, b > 0$, then $\text{lcm}(a, b) = ab/\gcd(a, b)$

Proof

- Let $d = \gcd(a, b)$, $a = dp$, $b = dq$
- Then $m = ab/d = pb = qa$ is a multiple of a and b
- If there was a smaller multiple $\bar{m} < m$, then $\bar{d} = ab/\bar{m} > d$ would be a common divisor: $a/\bar{d} = \bar{m}/b$, $b/\bar{d} = \bar{m}/a$

□

Outline

Least Common Multiple

Diophantine Equations: Examples

Diophantine Equations: Theorem

Modular Division

Diophantine Equations

A Diophantine equation is an equation where only integer solutions are allowed



Diophantine Equations: Examples

- 1 apple costs 22 pesos

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- The cashier only has 5-peso bills

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- $3x = 22 + 5y$, x, y are non-negative integers

Diophantine Equations: Examples

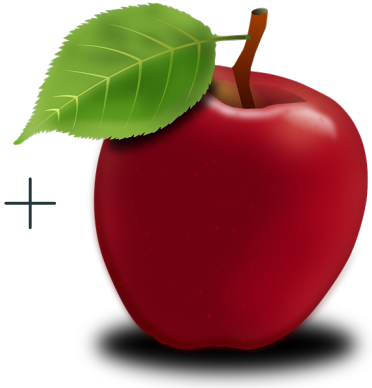
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$\times 9 =$



$\times 1 +$



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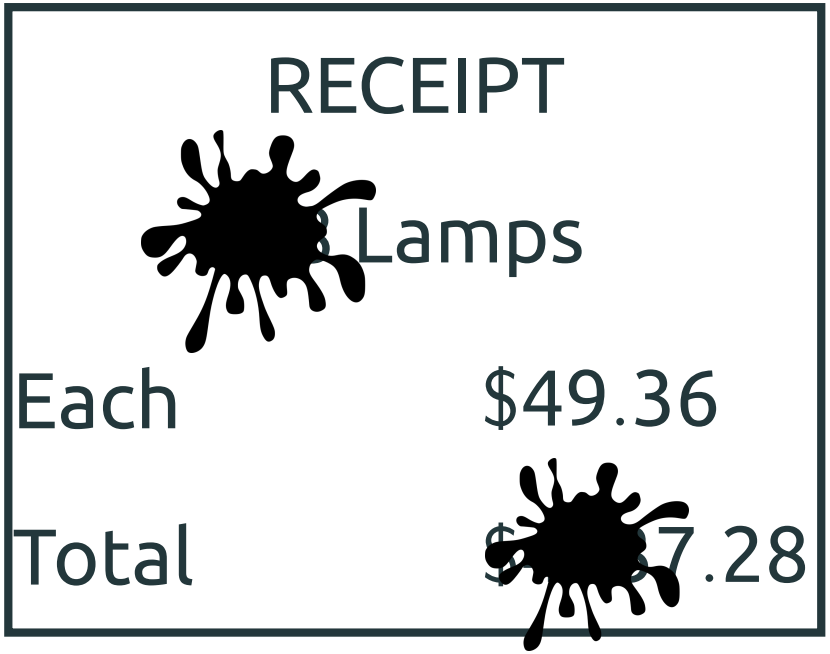
$\times 14 =$



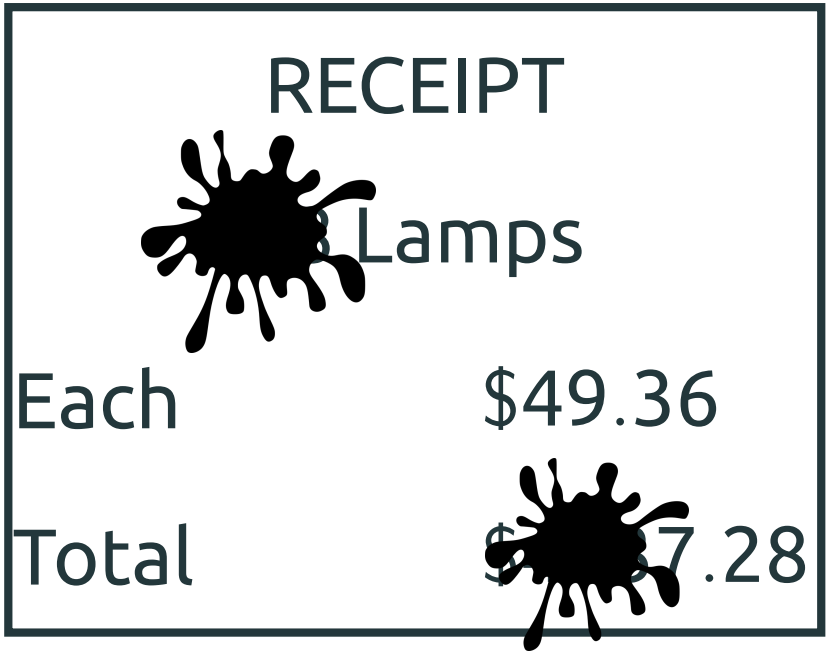
$\times 4 +$



Diophantine Equations: Examples

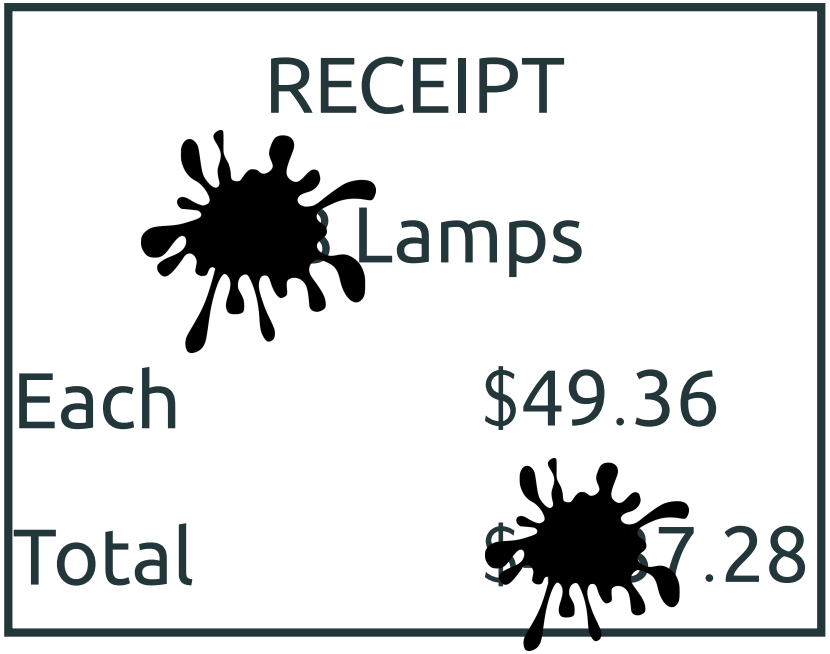


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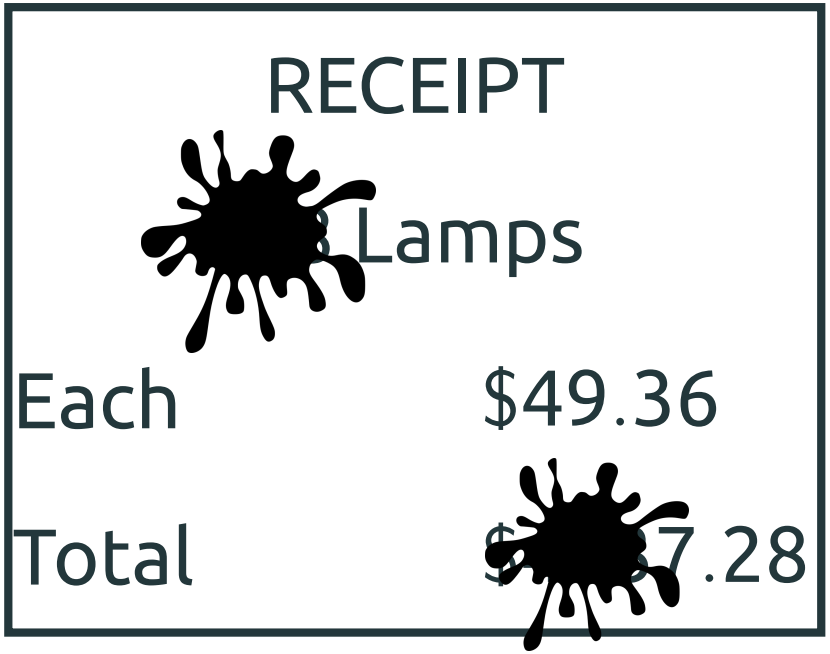
- *x* lamps, each 4936 ¢

Diophantine Equations: Examples



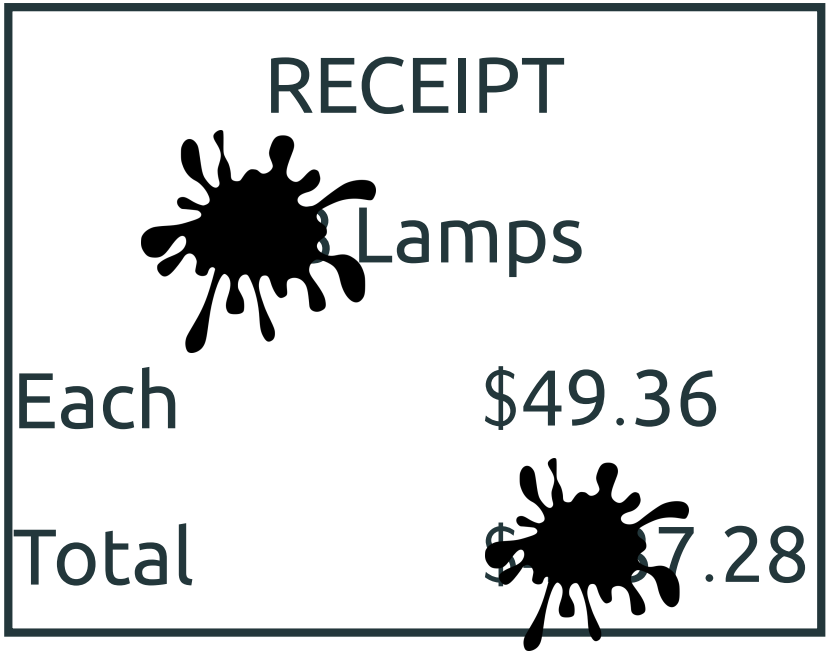
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Diophantine Equations: Examples



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- Total: $1000y + 728$ ¢

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- $4936x = 1000y + 728$, x, y are non-negative integers, $100 \leq y < 1000$

Diophantine Equations: Examples

RECEIPT	
98 Lamps	
Each	\$49.36
Total	\$4837.28

- x lamps, each 4936 ¢
- $100 \leq y < 1000$
- Total: $1000y + 728$ ¢
- $4936x = 1000y + 728$, x, y are non-negative integers, $100 \leq y < 1000$

Diophantine Equations: Examples

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  - No solutions!

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  - $187 \cdot (-2) + 55 \cdot 9 = 121$
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When does a Diophantine equation have solutions?

Outline

Least Common Multiple

Diophantine Equations: Examples

Diophantine Equations: Theorem

Modular Division

Solutions of Diophantine Equations

Theorem

Given integers a, b, c (at least one of a and $b \neq 0$), the Diophantine equation

$$ax + by = c$$

has a solution (where x and y are integers) if and only if

$$\gcd(a, b) \mid c\,.$$

Proof of Theorem

Proof

Let $d = \gcd(a, b)$

Proof of Theorem

Proof

$$\begin{aligned} & \text{Let } d = \gcd(a, b) \\ \Rightarrow\ & a = dp \text{ and } b = dq, \text{ thus,} \\ & c = ax + by = d(px + qy) \end{aligned}$$

Proof of Theorem

Proof

Let $d = \gcd(a, b)$

$\Rightarrow$ $a = dp$ and $b = dq$, thus,
$c = ax + by = d(px + qy)$

$\Leftarrow$ Extended Euclid's algorithm:
$a\bar{x} + b\bar{y} = d$

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$\Rightarrow$ $a = dp$ and $b = dq$, thus,
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$\Leftarrow$ Extended Euclid's algorithm:
$a\bar{x} + b\bar{y} = d$

If $c = td$, then $x = t \cdot \bar{x}, y = t \cdot \bar{y}$:
$ax + by = t(a\bar{x} + b\bar{y}) = td = c$ $\square$

Finding a Solution

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  - $-69 = 391 \cdot (-3) \cdot (-3) + 299 \cdot 4 \cdot (-3)$
  - $x = 9, y = -12$
  - But $x = -4, y = 5$ is also a solution. How do we find all solutions?

Euclid's Lemma

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If $n \mid ab$ and $\gcd(a, n) = 1$, then $n \mid b$.

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Proof

- From Extended Euclid's algorithm (a, n):
- $ax + ny = 1$
- $axb + nyb = b$
- From $ab = kn$,
  $b = axb + nyb = n(xk + yb)$ $\square$

Finding All Solution

Theorem

Let $\gcd(a, b) = d, a = dp, b = dq$. If (x_0, y_0) is a solution of the Diophantine equation $ax + by = c$:

$$ax_0 + by_0 = c\,,$$

then all the solutions have the form

$$a(x_0 + tq) + b(y_0 - tp) = c\,,$$

where t is an arbitrary integer.

Proof of Theorem

Proof

- $a = dp, b = dq, ax_0 + by_0 = c$

Proof of Theorem

Proof

- $a = dp, b = dq, ax_0 + by_0 = c$
- For any integer t,

$$\begin{aligned} & a(x_0 + tq) + b(y_0 - tp) \\ &= ax_0 + by_0 + t(aq - bp) \\ &= c + t(dpq - dpq) = c \end{aligned}$$

is a solution

Proof of Theorem

Proof (continued)

- Consider 2 solutions: (x_1, y_1) and (x_2, y_2)

Proof of Theorem

Proof (continued)

- Consider 2 solutions: (x_1, y_1) and (x_2, y_2)
- $a(x_1 - x_2) + b(y_1 - y_2) = c - c = 0$

Proof of Theorem

Proof (continued)

- Consider 2 solutions: (x_1, y_1) and (x_2, y_2)
- $a(x_1 - x_2) + b(y_1 - y_2) = c - c = 0$
- $p(x_1 - x_2) + q(y_1 - y_2) = 0$

Proof of Theorem

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- $\gcd(p, q) = 1$

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- $\gcd(p, q) = 1$
- By Euclid's lemma: $x_1 - x_2 = tq$

Proof of Theorem

Proof (continued)

- Consider 2 solutions: (x_1, y_1) and (x_2, y_2)
- $a(x_1 - x_2) + b(y_1 - y_2) = c - c = 0$
- $p(x_1 - x_2) + q(y_1 - y_2) = 0$
- $\gcd(p, q) = 1$
- By Euclid's lemma: $x_1 - x_2 = tq$
- Then $y_1 - y_2 = -tp$ $\square$

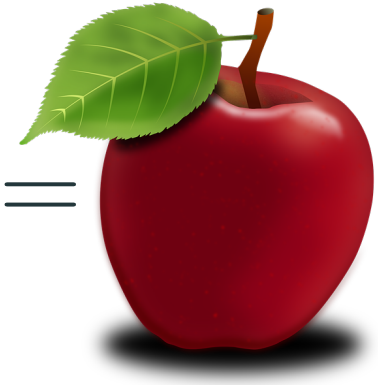
Example


BANCO CENTRAL DE CUBA
902859
902859
3
3
PESOS
FA-34
FA-34

×9 +


BANCO CENTRAL DE CUBA
145018
145018
5
CINCO
PESOS
EC-15
EC-15
AÑO 2000
PESOS

×1 =



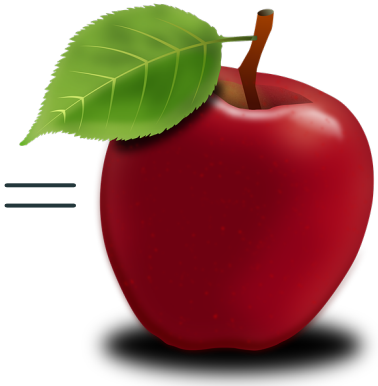
Example



$\times 9 +$



$\times 1 =$



- $3x + 5y = 22$

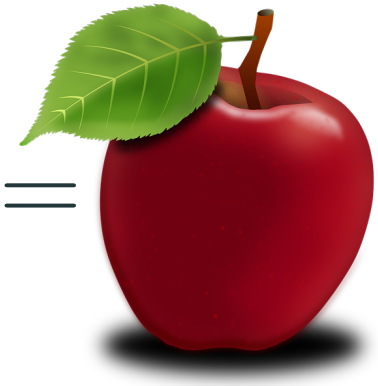
Example



$\times 9 +$



$\times 1 =$



- $3x + 5y = 22$
- $x_0 = 9, y_0 = -1$

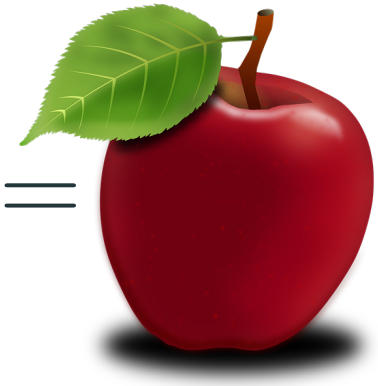
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- $3x + 5y = 22$
- $x_0 = 9, y_0 = -1$
- $a = 3, b = 5, d = 1$

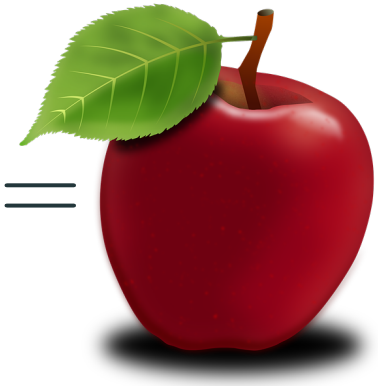
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- $a = 3, b = 5, d = 1$
- $a = dp, b = dq, p = 3, q = 5$

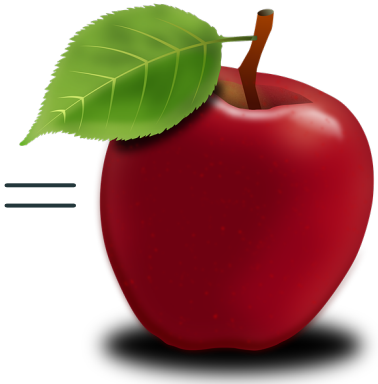
Example



$\times 9 +$



$\times 1 =$



- $3x + 5y = 22$
- $x_0 = 9, y_0 = -1$
- $a = 3, b = 5, d = 1$
- $a = dp, b = dq, p = 3, q = 5$
- All solutions:

$$x = x_0 + tq = 9 + 5t$$

$$y = y_0 - tp = -1 - 3t$$

Example

- All solutions:

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Example

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$$y = y_0 - tp = -1 - 3t$$

- If we want $x \geq 0$ and $y \leq 0$, then take

$$9 + 5t \geq 0$$
$$-1 - 3t \leq 0$$

Example

- All solutions:

$$x = x_0 + tq = 9 + 5t$$
$$y = y_0 - tp = -1 - 3t$$

- If we want $x \geq 0$ and $y \leq 0$, then take

$$9 + 5t \geq 0$$
$$-1 - 3t \leq 0$$

- That is, $t \geq -1/3$, or $t \geq 0$

Outline

Least Common Multiple

Diophantine Equations: Examples

Diophantine Equations: Theorem

Modular Division

Division mod 7

×	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
5	0	5	3	1	6	4	2
6	0	6	5	4	3	2	1

Division mod 7

$\times$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
5	0	5	3	1	6	4	2
6	0	6	5	4	3	2	1

- Given $a \neq 0$ and b, there exists x such that $a \times x \equiv b \pmod 7$

Division mod 7

$\times$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
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- Given $a \neq 0$ and b, there exists x such that $a \times x \equiv b \pmod 7$
- x plays the role of modular division $x = b/a \pmod 7$

Division mod 6

×	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Division mod 6

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

- $2/5 \equiv 4 \pmod 6$.
  Indeed, $4 \times 5 \equiv 2 \pmod 6$

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- $2/5 \equiv 4 \pmod 6$.
  Indeed, $4 \times 5 \equiv 2 \pmod 6$
- But there is no x s.t. $3 \times x \equiv 1 \pmod 6$
- We can't divide 1 by 3 modulo 6!

Multiplicative Inverse

- A multiplicative inverse of $a \bmod n$ is $\bar{a}$ s.t.

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- Indeed, for every b,

$$b/a \times a \equiv b \times \bar{a} \times a \equiv b \pmod{n}$$

Uniqueness of Inverses

Lemma

If a has a multiplicative inverse, then it is unique

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Proof

If x and y are multiplicative inverses of a, then

$$x = x \times (a \times y) = (x \times a) \times y = y$$

□

Existence of Inverses

Theorem

a has a multiplicative inverse modulo n if and only if $\gcd(a, n) = 1$

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Proof

- $ax \equiv 1 \pmod{n}$ iff $ax + kn = 1$
- For fixed a and n, this Diophantine equation has a solution (x) iff $\gcd(a, n) \mid 1$

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  - s is the multiplicative inverse of a modulo n
  - Now $x \equiv b/a \equiv b \times s \pmod{n}$

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